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A Cross-Layer SDN Orchestration Architecture for Stateful Edge Services

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Master's in Telecommunications and Computer Engineering

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Ph.D. Professor Pedro Ramos, Assistant Professor,
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Month, Year

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TECHNOLOGY
AND ARCHITECTURE

Department of Information Science and Technology

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Acknowledgments

Write here your acknowledgements and, if applicable, any grants or sources of funding.

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Resumo

Palavras-Chave:

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Abstract

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Keywords: *Keywords; separated; by; semicolons*

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Figure 1.1 Global internet traffic between 2020 and 2025. Source: Statista, based on ITU data [1].

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List of Acronyms

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CHAPTER 1

Introduction

Studies have shown that edge web services encompassing routing, monitoring and auto-scaling can suffer from coordination delays that arise from the independent control loops through which each module operates. This accumulation of handoff delays between components degrades service quality and can reduce orchestration reliability. In this chapter the context, motivation and problem statement are explained in 1.1 with the objectives presented in 1.2. The research questions that emerged are available in 1.3. The methodology used to develop the thesis is detailed in 1.4, the main contributions in 1.5 and finally the outlining of the rest of the 7 chapters of the thesis are presented in 1.6.

1.1. Context, Motivation, and Problem Statement

In the year 2025 global internet traffic surpassed 7 exabytes (1 exabyte = 1 000 000 000 gigabytes), this corresponds to a more than double increase in the amount of internet traffic generated since 2020 across fixed and mobile networks, this is represented in Figure 1.1. [1].

This growth has been driven by many technological advancements some of which, have in turn galvanized the emergence of real-time interactive and data-intensive applications among others. Examples include web streaming and cloud gaming, which are both services where users expect sub-second latency and high resolution multimedia, highlighting the need for fast and capable data processing. These are just a few examples that demonstrate the demand of infrastructure that is able to deliver performance with high availability and reliability at global scale, in other words regardless of where the end users are located.

Cloud computing is and has been the [2] backbone that enables these applications to operate at a global scale. It offers automatic scaling, both horizontally and vertically, a pay-per-use model and among other aspects, [2] a globally distributed offering and availability that has established it as the foundational layer that supports modern applications and services.

But at the same time, a growing percentage of global internet traffic comes from mobile devices, which now represent the main gateway to internet access [3]. This ongoing shift underscores that the majority of internet access comes from the network edge, far from the cloud data centers, evidencing the importance of edge computing as an extension or alternative to cloud computing as it enables geo-distributed deployments of applications and services. As presented in [4], [5] edge computing brings the computational resources to the network edge which and therefore closer to the end-users and end-products allowing for much lower latency in provisioning services and applications to users. Beyond lower

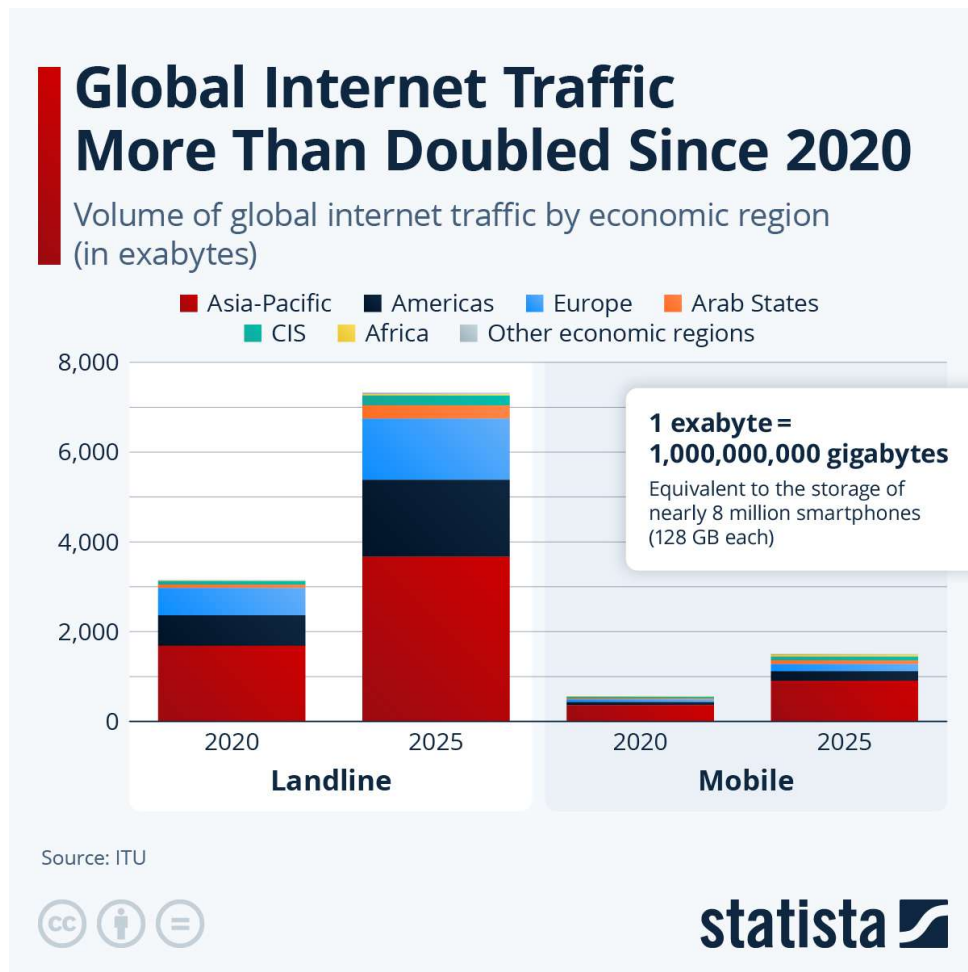


FIGURE 1.1. Global internet traffic between 2020 and 2025. Source: Statista, based on ITU data [1].

latency, edge computing’s inherent geo-distributed servers dispersed geographically and close to users relieves bandwidth pressure on the core network, since data can be processed and stored locally, at the edge, for quicker responses, only a subset of it is sent to the cloud [4], freeing up backbone available bandwidth.

Despite these advantages, this geographical distribution also translates into less capacity per location and greater complexity in managing the distributed edge servers. As Satyanarayanan puts it, “the dispersion inherent in edge computing raises the complexity of management considerably” [5]. This complexity stems from the properties that define the edge itself: its resources are dispersed across many sites, user demand varies considerably over time, and the requests served are heterogeneous in their resource requirements. Luo et al. further highlight that, without efficient resource scheduling, this distributed nature leads to resource waste [6]. Resource provisioning therefore becomes increasingly important, since the limited capacity of each location must be adjusted over time to suit the constant fluctuation of user demand.

The resource management problem is compounded when the services themselves are stateful. Unlike stateless microservices, which can be replicated freely and absorb traffic

immediately, stateful services in a geo-distributed edge deployment depend on data co-located with compute: a content item ingested at one site is served from the database of that site, so a request reaching a server elsewhere must either be served remotely, paying WAN latency on every access; or wait for the hot data to be replicated locally, paying synchronisation overhead; or require a full replica, paying storage and replication cost [7], [8], [9]. Scale-out alone does not solve this: the orchestration system must detect when data must be brought closer, deliver that information to the decision point, provision the right kind of capacity, and redirect traffic to it — and each of these steps introduces delay that directly degrades service quality under the time pressure of an ongoing demand shift.

This thesis uses a Multi-Region Content Discovery Platform — a geo-distributed edge deployment — as its experimental workload: content items are ingested regionally and discovered globally through tag-based personalized feeds, with heterogeneous document types and two stress regimes — one driven by data locality, the other by compute-analytics throughput.

1.2. Objectives

1.3. Research Questions

1.4. Research Methodology

1.5. Main Contributions of this Dissertation

1.6. Dissertation Structure

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CHAPTER 2

Background and Related Work

- 2.1. Review Methodology
- 2.2. The Unexamined Default: Three-Layer Separation
- 2.3. Auto Scaling
- 2.4. Load Balancing on SDN
- 2.5. Monitoring & Telemetry
- 2.6. Resource Orchestration on SDN
- 2.7. Summary and Research Gaps

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CHAPTER 3

Architecture and Design of Proposed Solution for Edge Services

This chapter presents the design of the proposed cross layer sdn orchestration architecture for edge services. It stems from the research gaps explored in the 2 and brings to the fold the need for an architecture that allows for each mentioned component to be independently managed and tested.

3.1. Design Requirements

3.2. Architecture

3.3. Distributed Elastic Allocation of Resources

3.3.1. Services

3.3.2. Data

3.4. Monitoring and Decision Engine

3.5. Control Workflow

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CHAPTER 4

Implementation of Proposed Solution

4.1. Experimental infrastructure

4.2. Distributed Elastic Allocation of Resources

4.2.1. Services

4.2.2. Data

4.3. Monitoring and Decision Engine

4.4. Control Workflow

4.5. Implementation Validation

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CHAPTER 5

Evaluation Methodology

5.1. Experimental Objectives

5.2. Evaluation Scenarios

5.3. Performance Metrics

5.4. Experimental Procedure with Results Statistical Analysis

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CHAPTER 6

Experimental Results

- 6.1. Telemetry Delivery Semantics (RQ1)
- 6.2. Bottleneck-Aware Scaling Action (RQ2)
- 6.3. Readiness Propagation and Traffic Admission (RQ3)
- 6.4. Network Performance
- 6.5. Scalability Analysis
- 6.6. Discussion: Synthesis Across Interfaces

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CHAPTER 7

Conclusions and Future Work

7.1. Conclusions

7.2. Research Contributions

7.3. Limitations

7.4. Future Research Directions

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